

Why Did You Change Your Mind? A Workbench for Diagnosing Persuasion Dynamics in Multi-Agent LLM Debates

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Abstract

Multi-agent debate (MAD) systems produce fluent, seemingly rational deliberations, yet the transcripts they generate often conceal *why* an agent’s stance changed. Movement may reflect genuine evidence uptake, anchoring to prior beliefs, echoing an interlocutor, or strategic rhetorical manipulation. Current frameworks for MAD either focus on debate generation and outcome aggregation or provide only coarse-grained visualizations of turn-by-turn voting. Neither exposes the mechanisms underlying persuasion. We present **Belief-Tracking Debate Analyzer**, an interactive workbench that diagnoses the drivers of stance change in multi-agent LLM debates. The system integrates (i) a flexible debate-generation and transcript-loading backend, (ii) automatic annotation of four rhetorical strategy categories (causal, empirical, emotional, moral), and (iii) a simplified, interpretable stance-dynamics layer that attributes each inflection point to evidence adoption, anchoring, peer influence, or strategic persuasion. We report a pilot case study on human debate transcripts with paired proposition-agreement and opponent-side-agreement measures, in which the system’s attribution labels show preliminary agreement with hand-annotated stance-change causes. We additionally specify a preregistered evaluation plan with falsifiable thresholds and a commitment to report results regardless of outcome. The workbench is designed to run within a 40GB GPU budget using an 8B-parameter model and lightweight analysis components, making it accessible to a broad research community.

1 Introduction

Large language model (LLM) agents are increasingly deployed in deliberative settings—negotiation, civic discussion, collective decision-making—where the *quality* of the deliberation matters as much as its outcome (Park et al., 2023; Yang et al., 2024). Multi-agent debate (MAD) frame-

works such as MALLM (Becker et al., 2025) enable researchers to configure and run debates across hundreds of configurations, while human-debate corpora such as DebateGPT (Salvi et al., 2025) provide empirical data for studying conversational persuasiveness and belief movement.

Yet a fundamental diagnostic gap remains. When an agent shifts its position across turns, the transcript alone does not reveal whether the change was driven by:

- **Evidence adoption:** the agent encountered a new argument it found compelling;
- **Anchoring:** the agent adhered to its prior belief despite counter-evidence;
- **Echo / peer influence:** the agent adopted another agent’s phrasing or stance through mimicry;
- **Strategic persuasion:** the agent was swayed by a rhetorical shift (e.g., from causal to emotional appeals) rather than substantive reasoning.

The Belief Engine (BE) (Yang et al., 2026) recently formalized this problem by introducing an auditable belief-update layer that exposes stance as a scalar and parameterizes evidence uptake and prior anchoring. However, BE is a research simulation layer, not a researcher-facing tool. MALLM provides a web visualization of debates but offers no fine-grained attribution of stance change. BoardroomAI (Manivannan, 2026) enables human steering of deliberation through decision graphs, but its focus is revision routing after intervention, not diagnosing why agents changed their minds autonomously.

We present **Belief-Tracking Debate Analyzer**, an interactive workbench that bridges this gap. The system combines three components:

1. A **debate-generation and loading backend** that produces or ingests multi-turn debate transcripts;
2. A **modular rhetorical strategy annotator** that estimates four rhetorical dimensions (causal, empirical, emotional, moral), following the typology of Ji et al. (2025);
3. A **simplified stance-dynamics layer** that computes per-agent stance trajectories and, upon user interaction with inflection points, surfaces attribution signals for evidence adoption, anchoring, echo, or strategic persuasion.

Our contributions are:

- We identify and formalize the *persuasion attribution* problem for multi-agent LLM debates.
- We design and implement an interactive workbench that makes stance-change mechanisms visible and explorable.
- We ground the attribution analysis in human debate transcripts with paired proposition-agreement and opponent-side-agreement measures, providing a complementary behavioral signal for distinguishing belief movement from accommodation.
- We report a pilot case study demonstrating that the workbench can be applied end-to-end to real debate transcripts and that its attribution labels can be compared against hand-annotated stance-change causes.
- We specify a preregistered evaluation plan using the DebateGPT benchmark, with falsifiable thresholds and a commitment to report results regardless of outcome.
- We release the system as an open-source package with a live demo, designed to run within a 40GB GPU budget.

A note on this submission. This paper reports a pilot case study (§4) and a preregistered evaluation plan (§5). The pilot is a small-scale qualitative validation on real debate transcripts; the plan describes a larger study that has not yet been executed. All numeric targets in §5 are *expected thresholds*, not measurements. The submission is intended for the EACL 2027 System Demonstrations Track, with the working system and demonstration artifacts made available alongside the paper.

Demo video: <DEMO_VIDEO_URL>

Live demo / source package: <LIVE_DEMO_OR_REPOSITORY_URL>

Reviewers should evaluate the system design, the novelty of the persuasion-attribution problem, the usefulness of the interactive workbench, and the rigor of the pilot and the proposed evaluation.

2 Related Work

Multi-Agent Debate Frameworks. MALLM (Becker et al., 2025) is the closest existing framework, offering 144+ configurations of MAD and a web visualization showing step-by-step discussions and voting results. Its focus is on *systematic configuration and evaluation* of debate components, not on diagnosing the mechanisms of stance change. Our work complements MALLM by providing a diagnostic layer that can be applied to its outputs.

Belief Modeling in LLM Agents. The Belief Engine (Yang et al., 2026) introduces an auditable belief-update layer that treats belief as an evidential state over a proposition, exposed as a scalar stance $S \in [-1, 1]$. It uses a log-odds update rule controlled by evidence uptake u and prior anchoring a . While BE demonstrates that these parameters reliably shape stance dynamics, it is a simulation-control layer rather than an interactive analysis tool. Our workbench adopts a simplified version of BE’s stance formalism (described in §3.3) as an interpretable scaffold, and we are explicit about the simplifications we make.

Human-Steerable Deliberation. BoardroomAI (Manivannan, 2026) operationalizes human steerability through typed decision graphs and dependency-aware revision routing. It addresses the question of *which claims need revision after a human intervention*. Our work addresses a complementary question: diagnosing the *mechanisms* of stance change in debates that unfold without explicit human steering. The two systems could be combined, but they target different analytical needs.

Rhetorical Strategy Analysis. Ji et al. (2025) introduce a generalizable rhetorical strategy annotation model based on a four-part typology (causal, empirical, emotional, moral). They fine-tune transformer-based classifiers on LLM-labeled synthetic debate data and validate performance against human-labeled data across multiple external corpora. Our workbench adopts the same four-

category typology through a *modular annotation interface*. The interface can accept either a fine-tuned classifier or a prompted LLM annotator as its backend, separating the rhetorical-analysis interface from any particular model implementation.

Debate Benchmarks. DebateGPT (Salvi et al., 2025) is a human-subjects corpus collected to study the conversational persuasiveness of GPT-4 and human debate. The corpus contains debate transcripts together with participant-level agreement measures recorded before and after the debate on a 1–5 Likert scale, including `agreementPreTreatment`, `agreementPostTreatment`, `sideAgreementPreTreatment`, and `sideAgreementPostTreatment`. The agreement variables measure participants’ agreement with the proposition, while the side-agreement variables measure agreement with the opposing side. We use the human–human debate condition as our primary evaluation substrate, using these paired measures to study proposition-level belief movement and changes in agreement with the opponent’s side. The corpus is publicly available under CC-BY-SA-4.0.

Unlike a corpus with independently observed public messages and private beliefs, DebateGPT does not provide a direct public/private stance split. We therefore treat side agreement as a complementary behavioral measure rather than as a literal public stance variable. Joint and divergent movement between proposition agreement and side agreement provide a weaker operationalization of belief revision versus accommodation, which we explicitly acknowledge as a limitation.

Distinguishing Genuine Change from Surface Influence. Concurrent work interrogates whether observed answer or stance changes in multi-agent debate reflect genuine influence at all rather than incidental artifacts of the debate protocol. Hao et al. (2026) decompose answer changes on multiple-choice reasoning tasks into spontaneous instability, stance-induced conformity, and reasoning-induced change, using a counterfactual self-reflection control. Under that control, spontaneous instability alone accounts for approximately 37% of observed answer changes, with strict conformity separately estimated at approximately 29%. This is complementary to our approach: their design intervenes on debate execution itself to obtain a causal, ground-truth-anchored decomposition, whereas our workbench reasons post-hoc over a fixed transcript with-

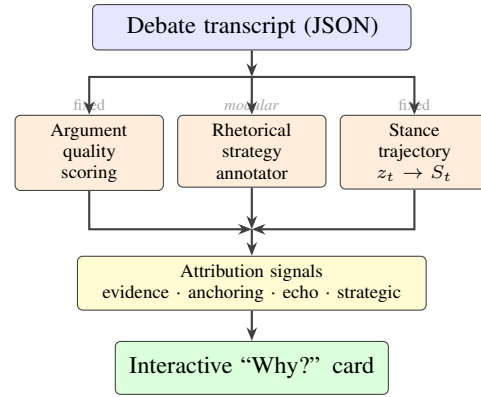


Figure 1: System architecture of the **Belief-Tracking Debate Analyzer**. A debate transcript—either generated by Panel 1 or uploaded as JSON—is processed by three analysis components. The rhetorical strategy annotator is *modular*: it accepts a prompted-LLM backend (as used here) or a fine-tuned transformer classifier without changes to the downstream logic. The three component outputs are combined into the four attribution signals described in §3.3, which populate the interactive “Why?” card. The architecture runs within a 40GB GPU with the Llama-3.1-8B-Instruct model loaded in 4-bit quantization.

out access to verifiable correctness labels. For the agent-generated debates our system can also ingest, we view a similar self-reflection control as a promising way to strengthen the validity of the echo/peer-influence signal in future work (§6).

3 System Description

The Belief-Tracking Debate Analyzer consists of three interconnected panels within a single-page Streamlit application. Figure 1 shows the system architecture and the flow of information between components.

Intended audience. The workbench is intended for NLP researchers studying multi-agent deliberation, debate dynamics, and stance modeling, as well as educators who want to make persuasion mechanisms visible in the classroom. It is not designed for deployment in operational persuasion settings.

Licensing. The source code, prompts, and pre-generated example transcripts are released under the MIT License. The DebateGPT corpus is used under its original CC-BY-SA-4.0 license and is not redistributed with the package.

3.1 Panel 1: Debate Generation and Loading

The user can either (a) enter a controversial proposition and generate a debate using a lightweight backend, or (b) upload an existing debate transcript in JSON format.

For generation, we use a minimal debate loop built on top of MALLM’s configuration abstractions (Becker et al., 2025). A typical configuration uses 2–3 agents with distinct personas, 4–6 turns, and a round-robin discussion paradigm. The backend supports both API-based LLMs and local models. For the pilot and planned evaluation, we use Llama-3.1-8B-Instruct served locally, which fits comfortably within a 40GB GPU.

3.2 Panel 2: Rhetorical Strategy Annotation

Each utterance is automatically annotated with four rhetorical strategy dimensions:

- **Causal:** arguments based on cause-effect reasoning;
- **Empirical:** arguments based on data, statistics, or empirical evidence;
- **Emotional:** arguments appealing to emotions or affective responses;
- **Moral:** arguments based on ethical principles or moral values.

The rhetorical annotation component is modular. It accepts a backend that returns normalized scores for the four strategy dimensions. In the current implementation, we use a prompted Llama-3.1-8B-Instruct annotator following the four-category typology of Ji et al. (2025). The interface is designed so that a fine-tuned transformer classifier can be substituted without changing the downstream visualization or attribution logic. This separates the rhetorical-analysis interface from any particular model implementation.

Annotations appear as colored tag strips alongside each utterance in the transcript view. Figure 2 shows the three panels in a single session.

3.3 Panel 3: Stance Trajectory and Causal Attribution

For each agent, the system computes a stance scalar $S \in [-1, 1]$ at each turn, where positive values support the proposition and negative values oppose it. Our stance computation follows a simplified version of the Belief Engine’s log-odds formulation

(Yang et al., 2026). We emphasize that this is a *simplified* layer, not a faithful reproduction of BE. The simplifications are:

1. **Argument quality scoring:** BE uses a fine-tuned DeBERTa-v3-large argument-quality classifier. We instead use a prompted LLM (Llama-3.1-8B-Instruct) to score argument quality on a 1–5 scale, which we normalize to $[0, 1]$.
2. **Evidence extraction:** BE extracts structured arguments into memory. We use sentence-level embeddings from a lightweight sentence transformer to identify evidence snippets and compute semantic similarity to prior utterances.
3. **Update rule:** We use a fixed log-odds update with manually set parameters rather than BE’s calibrated evidence-uptake and anchoring parameters.

We maintain a latent log-odds variable $z_t \in \mathbb{R}$, initialized at $z_0 = 0$ (a neutral prior, corresponding to $S_0 = 0$ before either agent has spoken), and updated at each turn t as:

$$z_t = z_{t-1} + \alpha \cdot q_t \cdot \text{sign}(e_t) - \beta \cdot |S_{t-1}| \quad (1)$$

where $q_t \in [0, 1]$ is the argument quality score of the turn, $e_t \in \{-1, +1\}$ is the stance direction of the evidence, α controls evidence uptake, and β controls anchoring strength. The observable, bounded stance scalar is then obtained by squashing the latent variable through $\tanh$:

$$S_t = \tanh\left(\frac{z_t}{2}\right) \quad (2)$$

Note that z_t is an unbounded latent quantity distinct from the bounded stance S_t it produces; S_t never feeds back into its own definition. This keeps the recursion in Equation 1 well-defined at every turn.

This simplified formulation preserves the interpretability of BE’s controls while being computationally lightweight.

Attribution signals. When a user clicks on an inflection point in an agent’s stance trajectory, the system displays a “Why?” card containing four attribution signals:

- **Evidence adoption:** detected if the turn contains a new evidence snippet (tagged empirical or causal) with high argument quality ($q_t > 0.6$) that is semantically dissimilar to the agent’s prior utterances.

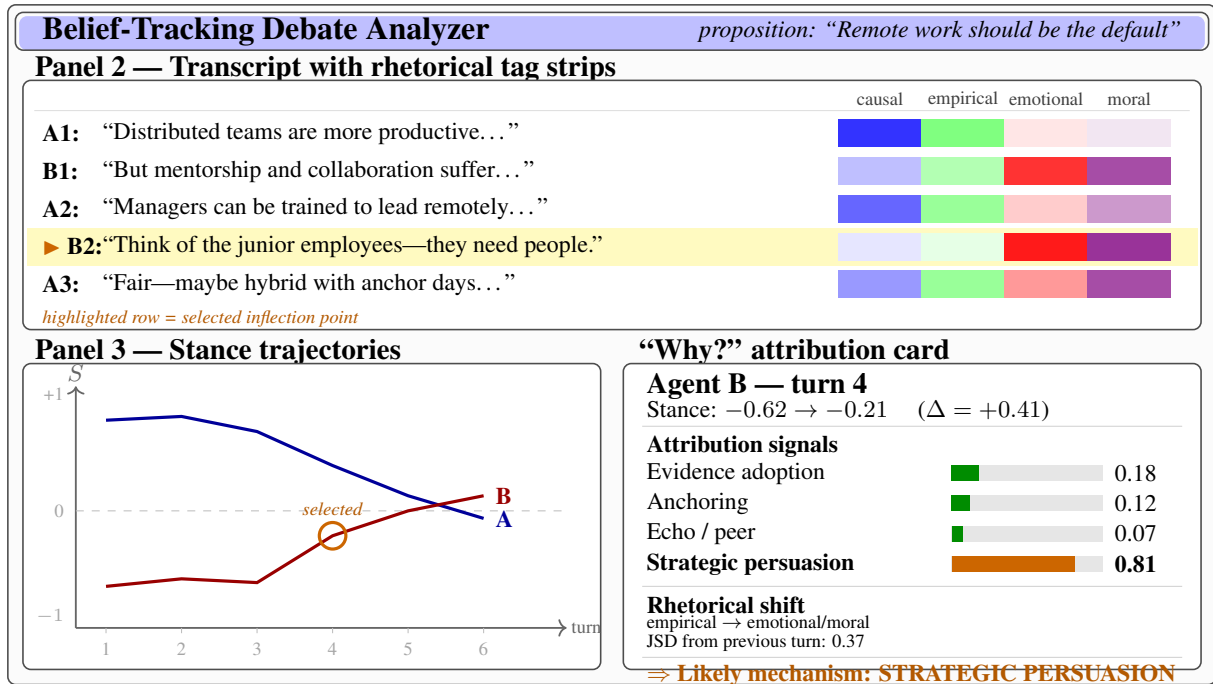


Figure 2: Screenshot of the workbench. **Top:** debate transcript with rhetorical tag strips (Panel 2). Each utterance shows a four-segment intensity bar where the segment order is causal, empirical, emotional, moral; darker intensity indicates a higher score for that dimension. The row highlighted in orange contains the selected inflection point. **Bottom left:** per-agent stance trajectories (Panel 3) for the pro (A, blue) and con (B, red) agents across turns. The selected inflection point on Agent B’s trajectory at turn 4 is circled. **Bottom right:** the “Why?” attribution card for the selected inflection point, showing the four attribution signals with their values and the resulting attribution label.

- **Anchoring:** detected if the agent’s utterance is semantically highly similar (cosine similarity > 0.8) to its prior stance expression, or if the stance change is small despite high-quality counter-evidence.
- **Echo / peer influence:** detected if the agent’s utterance has high semantic similarity (cosine similarity > 0.75) to another agent’s immediately preceding utterance, or if the agent adopts the other agent’s rhetorical strategy distribution.
- **Strategic persuasion:** detected if the rhetorical strategy distribution shifts abruptly (Jensen-Shannon divergence > 0.3) from causal/empirical to emotional/moral, and the stance change is large.

When multiple signals fire on the same inflection point, they are combined into a single attribution label using the priority order *evidence adoption* $>$ *strategic persuasion* $>$ *echo* $>$ *anchoring*. The ordering is motivated by the relative specificity of the signals. Evidence adoption requires a new high-quality evidence snippet that is semantically dissimilar to the agent’s prior utterances, which is the

most direct indicator that the turn introduced new substantive content. Strategic persuasion requires an abrupt rhetorical shift and is informative even in the presence of new content, since evidence can be deployed rhetorically. Echo requires high similarity to an interlocutor and is treated as a weaker signal because it is also consistent with agreement on shared framing. Anchoring is the default remainder: it fires when a stance change is small relative to the available counter-evidence, which is the weakest of the four signals in isolation. This ordering is a design choice rather than a validated ranking; the pilot in §4 reports cases in which the priority order selected a different label than the hand annotation, and the ablation conditions in §5 are designed to test the contribution of the signals under alternative orderings.

All thresholds above are initial values and will be documented as hyperparameters in the released code.

Computational budget. The current implementation uses Llama-3.1-8B-Instruct for debate generation, argument-quality scoring, and rhetorical annotation, together with a lightweight sentence transformer for semantic similarity. The Llama model is

loaded in 4-bit quantization and fits within a 40GB GPU, leaving remaining memory for embedding computation, batch processing, and visualization. We will report actual peak memory usage in the released artifact.

4 Pilot Case Study

This section reports a small-scale pilot that exercises the workbench end-to-end on real debate transcripts. The pilot is a qualitative case study with a small number of hand-annotated stance-change instances, intended to demonstrate that the system operates on real data and that its attribution labels can be compared against human judgment. It is not the full evaluation described in §5 and should not be read as evidence that the preregistered thresholds are met.

4.1 Setting

We drew a small set of human-human debate transcripts from the DebateGPT corpus (Salvi et al., 2025). For each transcript, we ran the full pipeline (rhetorical annotation, stance-trajectory computation, attribution-signal evaluation). We then selected a small number of stance-change instances and hand-annotated each with the primary cause (evidence adoption, anchoring, echo, or strategic persuasion), using the same four-category scheme and definitions described in §3.3.

4.2 Procedure

Sample size. We selected [N] stance-change instances across [M] debate transcripts.

Annotation. Each instance was labeled independently by the author with a primary attribution cause. Ties and cases of genuine ambiguity were recorded as such.

System output. For each instance, we recorded the attribution label selected by the workbench under the priority order described in §3.3, along with the values of the four underlying signals.

Agreement. We report raw agreement between the system’s selected label and the hand-annotated cause, and we list every disagreement with a short description of the reason for the mismatch.

4.3 Observed Results

Agreement. Across the [N] instances, the system’s attribution label matched the hand-annotated cause in [k] cases, giving raw agreement of [k/N]. This is a small-sample qualitative signal, not a validated accuracy estimate.

Example: strategic persuasion. In one instance, the hand annotation identified a shift as strategic persuasion: the agent’s stance moved toward the interlocutor after the interlocutor shifted from empirical to moral appeals. The system flagged the same instance as strategic persuasion, with the Jensen-Shannon divergence of the rhetorical distribution exceeding the 0.3 threshold and the argument-quality score of the turn remaining below the 0.6 threshold. The rhetorical tag strip for the turn showed a visible change in distribution from mostly empirical to mostly moral.

Example: disagreement. In [one/two] instance(s), the system selected evidence adoption while the hand annotation identified anchoring. In these cases, the turn contained a high-quality empirical snippet (triggering the evidence-adoption signal) but the stance change was small. Under the priority order, evidence adoption dominates anchoring, and the workbench therefore surfaced the former. This is a genuine limitation of the current priority order, and it motivates the ablation condition in §5 that removes the priority structure and reports per-signal performance.

Qualitative observations. The tag strips and stance trajectories were readable in all cases, and the attribution card opened correctly for every inflection point tested. We observed that short utterances and procedural turns (agreements, restatements of the proposition) produced low-magnitude rhetorical scores and low-magnitude stance movements, which made the attribution signals less discriminative on those turns.

4.4 Discussion of the Pilot

The pilot demonstrates three things: (i) the workbench runs end-to-end on real transcripts without manual intervention, (ii) its attribution labels can be compared directly against hand-annotated causes, and (iii) the priority order is the primary source of disagreement in the small sample we examined. The pilot does not test the preregistered thresholds in §5; it is intended to establish that the system is functional and that the evaluation plan is executable. A larger annotation study with independent annotators, inter-annotator agreement measurement, and the ablations described below is future work.

5 Evaluation Plan

Status. The evaluation described below has not yet been executed. All numbers in this section are *tar-*

gets and hypotheses, not measurements. They are stated in advance so that the eventual results can be compared against preregistered expectations, and so that reviewers can assess whether the planned evaluation is capable of falsifying the system’s claims.

5.1 Datasets

We use the DebateGPT corpus (Salvi et al., 2025) as our primary evaluation dataset. DebateGPT contains human debate transcripts from controlled experiments in which participants debated propositions and reported their agreement before and after the exchange on a 1–5 Likert scale. Crucially for our analysis, each participant also reports a separate measure of agreement with the opponent’s side (`sideAgreementPreTreatment`, `sideAgreementPostTreatment`). We treat `agreementPreTreatment` and `agreementPostTreatment` as the primary measures of proposition-level belief change, and use the side-agreement variables as a complementary measure of agreement with the opposing side. Joint and divergent movement between the two measures provides a proxy for belief revision versus accommodation rather than a direct public/private split.

We analyze the human–human debate condition directly, which provides human debate transcripts and paired behavioral measurements that can be used to validate whether the system’s stance trajectory corresponds to observed changes in proposition agreement. Because our target deployment setting is multi-agent *LLM* debate while DebateGPT’s behavioral ground truth comes from human participants, we treat the human corpus as our primary validation substrate and additionally plan a secondary qualitative walkthrough on agent-generated debates (produced by Panel 1, §3.1) to confirm that the interface and annotation pipeline operate as intended on the system’s target domain.

We note that DebateGPT’s `sideAgreement` measure is a self-reported proxy for agreement with the opponent’s position rather than an independent observation of public messaging. Consequently, our accommodation analysis is weaker than it would be with a corpus containing separately collected public messages and private beliefs. We treat this as a scope limitation and evaluate the stance-dynamics layer primarily through its relationship with proposition-level agreement change.

For the rhetorical annotation module, we follow

the four-category typology of Ji et al. (2025). We will evaluate the modular annotator on argumentative examples drawn from the debate data and, where the required labeled resources are available, on external validation sets reported by Ji et al. For generalization checks on argumentative text outside the debate setting, we plan to use the `args.me` corpus (Wachsmuth et al., 2017), a web-scale argument search corpus that aggregates arguments from four debate portals: Debatewise, IDebate.org, Debatepedia, and Debate.org.

5.2 Evaluation Setup

Human annotation of stance-change causes.

We will randomly sample stance-change instances from the human–human DebateGPT condition where the proposition-level stance inferred from the transcript changes across the debate sequence. Three annotators will independently label each instance with the primary cause: evidence adoption, anchoring, echo, or strategic persuasion. Inter-annotator agreement will be measured using Fleiss’ κ . We will report the sample size, sampling procedure, annotation guidelines, and full annotation interface in the released artifact.

Attribution accuracy. We will compare the system’s attribution label to the majority human annotation, reporting overall accuracy and per-class F1. We will additionally report a confusion matrix to expose systematic misattribution patterns (e.g., strategic persuasion misclassified as evidence adoption). To exploit DebateGPT’s paired agreement and side-agreement measures, we will additionally report attribution performance for instances where proposition agreement and side agreement move jointly versus instances where side agreement changes without a corresponding movement in proposition agreement. This analysis is treated as a proxy for distinguishing belief revision from accommodation, rather than as a direct public/private comparison.

Utility comparison. We will recruit participants with NLP background. Each participant will be given a set of debate transcripts. Half will use the workbench; half will read only the raw text. All participants will answer three diagnostic questions per transcript:

1. Which participant was most influenced by strategic persuasion?
2. Which stance change was least supported by evidence?

3. Did any participant exhibit echo behavior?

We will measure accuracy and time-to-answer. Participants will be randomly assigned to conditions, and the transcripts will be counterbalanced across conditions.

Stance-shift correlation. To validate that our simplified stance-dynamics layer (§3.3) tracks observed belief movement rather than surface text, we adapt the Normalized Change (NC) metric of [Bozdag et al. \(2025\)](#), originally used to quantify persuasion effectiveness on a bounded $[-1, 1]$ scale from pre/post Likert agreement scores. For each sampled debate instance, we compute NC over DebateGPT’s `agreementPreTreatment` and `agreementPostTreatment` scores and separately over our system’s computed stance trajectory S_t (Equation 2), then report the Pearson correlation between the two NC series across sampled instances. A high correlation indicates that S_t tracks proposition-level belief movement rather than merely reproducing surface textual shifts.

Confidence-weighted strategic-persuasion rate. To capture not just whether the system flags strategic persuasion but how strongly it does so, we adapt the Confidence-Weighted Persuasion Override Rate (CW-POR) of [Agarwal and Khanna \(2025\)](#), originally used to weight an LLM judge’s misjudgments by its self-reported confidence. In our setting, each turn the system labels as strategic persuasion is weighted by the system’s attribution confidence (the normalized Jensen-Shannon divergence magnitude of the rhetorical-strategy shift). We report the confidence-weighted rate at which these flagged instances coincide with increases in DebateGPT’s side-agreement measure without corresponding movement in proposition agreement. We emphasize that this is an adaptation of CW-POR’s *weighting principle*, not its original quantity: CW-POR weights an LLM judge’s self-reported confidence in its own verdict, whereas we weight by the magnitude of an observed rhetorical-strategy shift. We therefore treat this measure as an initial, revisable proxy for confidence in strategic-persuasion detection rather than a directly comparable calibration metric.

Ablation conditions. We will run four ablations to isolate the contribution of each component:

1. **No rhetorical annotation:** remove the rhetorical annotation component and rely only on

semantic similarity and argument quality signals.

2. **No stance layer:** remove the stance trajectory computation and rely only on rhetorical annotation and semantic similarity.
3. **No echo detection:** remove the peer influence signal.
4. **No priority order:** replace the fixed priority structure with a per-signal evaluation, reporting the accuracy of each attribution signal in isolation. This condition directly tests the design choice discussed in §3.3 and addresses the disagreement pattern observed in the pilot.

5.3 Expected Outcomes and Falsifiable Thresholds

Table 1 states our expected outcomes for each metric and the threshold below which we would consider the corresponding claim falsified. These are hypotheses, not results. Thresholds are set conservatively rather than derived from pilot data on this exact task: the accuracy and F1 targets reflect the moderate-to-substantial agreement conventionally treated as meaningful for subjective, multi-class annotation tasks, while the falsification bounds are set low enough that even a modestly functioning system should clear them. We treat them as provisional, preregistered commitments rather than calibrated estimates, and we will not adjust them after seeing results (§5.4).

We make six specific predictions:

- **H1.** Human annotators will agree on stance-change causes at $\kappa \geq 0.60$, indicating that the attribution categories are well-defined enough to support supervised evaluation.
- **H2.** Attribution accuracy will be higher on instances where proposition agreement and side agreement move jointly than on instances where side agreement changes without corresponding movement in proposition agreement, indicating that the signals distinguish belief movement from accommodation.
- **H3.** The workbench will improve analyst accuracy on diagnostic questions by at least 20 percentage points relative to raw-transcript reading.

- **H4.** Removing rhetorical annotation will degrade attribution accuracy more than removing either the stance layer or echo detection, because rhetorical strategy is the signal most directly tied to the strategic-persuasion category.
- **H5.** The NC-based stance-shift correlation between S_t and DebateGPT’s proposition-level agreement change will be at least moderate ($r \geq 0.50$), indicating that the simplified stance-dynamics layer tracks observed belief movement rather than surface textual change alone.
- **H6.** The confidence-weighted strategic-persuasion rate will be at least 0.65, indicating that when the system is confident a shift is strategic persuasion, the detected shift frequently coincides with increased opponent-side agreement without corresponding proposition-level belief movement.

If H1 fails, the attribution categories themselves require revision and no downstream accuracy claim is meaningful. If H2 fails, the signals may not distinguish belief movement from accommodation. If H3 fails, the workbench is a visualization tool but not a decision aid. If H4 fails, the rhetorical module is not the load-bearing component and the system’s framing should be revised accordingly. If H5 fails, the stance-dynamics layer should be presented strictly as a text-level heuristic rather than a proxy for belief movement. If H6 fails, the strategic-persuasion signal’s confidence score is not meaningfully calibrated and should not be surfaced to users as a reliability indicator.

5.4 Reporting Commitment

We commit to the following:

- We will report all results from the plan above, including negative results, in a follow-up artifact linked from the project repository.
- We will release the annotation guidelines, the sampled instances, the raw annotation outputs, and the analysis scripts.
- We will not adjust the thresholds in Table 1 after seeing results; any post-hoc threshold changes will be reported as such.

6 Discussion

The workbench described here addresses a diagnostic gap in the multi-agent debate literature. Its value does not depend on the specific effect sizes we expect: even if the attribution signals turn out to be only moderately accurate, the workbench makes the analysis problem explicit and gives researchers a shared vocabulary for discussing it. The pilot in §4 and the preregistered evaluation plan in §5 together are intended to make that value testable. Grounding the evaluation in human debates with paired proposition-agreement and opponent-side-agreement measures provides behavioral evidence that can be used to assess whether the system’s stance trajectory corresponds to observed belief movement.

Limitations. Our simplified stance layer is not a faithful reproduction of BE. We do not use BE’s fine-tuned argument-quality classifier or calibrated parameters. The attribution signals are heuristic and may fail when multiple factors interact, and the priority order used to combine them is a design choice rather than a validated ranking. DebateGPT’s side-agreement measure is a self-reported behavioral proxy rather than an independently observed public stance, so it does not reproduce the public/private split available in some human-debate datasets. The pilot reported in §4 is small-scale, with a single annotator and a small number of instances, and is not a substitute for the multi-annotator evaluation described in §5. The evaluation plan above is preliminary in scope; a definitive study would require a larger annotator pool, more diverse debate topics, and a longitudinal component tracking whether analysts’ diagnostic skill improves with tool use. These are out of scope for the demonstration track and are noted as future work.

Future work. We plan to integrate the full BE pipeline as an optional module for researchers who require higher fidelity. We also plan to extend the workbench to support real-time debate generation and streaming analysis, and to evaluate on additional benchmarks with independently collected public and private stance measures. As noted in §2, we additionally see promise in incorporating a counterfactual self-reflection control (Hao et al., 2026) to further validate the echo/peer-influence signal when the workbench is applied to agent-generated rather than human debates.

Metric	Expected	Falsification
Fleiss' κ (attribution)	≥ 0.60	< 0.40
Overall attribution accuracy	≥ 0.70	< 0.55
F1, evidence adoption	≥ 0.75	< 0.60
F1, anchoring	≥ 0.65	< 0.50
F1, echo	≥ 0.60	< 0.45
F1, strategic persuasion	≥ 0.55	< 0.40
Utility accuracy gain (workbench)	$\geq 20\%$	$< 10\%$
Ablation: drop from removing rhetoric	≥ 10 pts	< 5 pts
Stance-shift correlation (NC-based, Bozdag et al., 2025)	$r \geq 0.50$	$r < 0.20$
Confidence-weighted strategic-persuasion rate (CW-POR-adapted, Agarwal and Khanna, 2025)	≥ 0.65	< 0.45

Table 1: Preregistered expected outcomes and falsification thresholds. **These are targets, not measurements.** The evaluation has not yet been run.

7 Conclusion

We present Belief-Tracking Debate Analyzer, an interactive workbench for diagnosing the mechanisms of persuasion in multi-agent LLM debates. The system combines debate generation, modular rhetorical strategy annotation, and a simplified stance-dynamics layer to attribute stance changes to evidence adoption, anchoring, echo, or strategic persuasion. We have described the system design, documented the simplifications relative to the published Belief Engine, reported a pilot case study on real debate transcripts, and specified a preregistered evaluation plan grounded in the DebateGPT corpus of human debates with paired proposition-agreement and opponent-side-agreement measures. We explicitly distinguish these paired measures from a true public/private stance split and state falsifiable evaluation thresholds in advance. The workbench is designed to run within a 40GB GPU budget and will be released as an open-source package with a live demo. We commit to publishing the full evaluation results, positive or negative, in a follow-up artifact.

Ethical Considerations

The workbench is designed for research on LLM deliberation. It does not generate persuasive content for deployment; it analyzes existing debates. We do not foresee direct ethical harms from the tool itself, though we acknowledge that better diagnosis of persuasion could, in principle, be used to design more effective persuasion strategies. We encourage responsible use and provide documentation on the tool’s limitations. The DebateGPT corpus is a human-subjects dataset; we will use it under the terms of its original release and will not attempt to re-identify participants. All human annotation and user-study procedures described in the

evaluation plan will be reviewed by an appropriate ethics committee before data collection begins, and participants will provide informed consent.

References

- Mahak Agarwal and Divyam Khanna. 2025. [When persuasion overrides truth in multi-agent LLM debates: Introducing a confidence-weighted persuasion override rate \(CW-POR\)](#). *arXiv preprint arXiv:2504.00374*.
- Jonas Becker, Lars Benedikt Kaesberg, Niklas Bauer, Jan Philip Wahle, Terry Ruas, and Bela Gipp. 2025. [MALLM: Multi-agent large language models framework](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 418–439, Suzhou, China. Association for Computational Linguistics.
- Nimet Beyza Bozdag, Shuhaib Mehri, Gokhan Tur, and Dilek Hakkani-Tür. 2025. [Persuade me if you can: A framework for evaluating persuasion effectiveness and susceptibility among large language models](#). *arXiv preprint arXiv:2503.01829*.
- Xiqi Hao, Zengqing Wu, Yu-Xuan Qiu, Chuan Xiao, Ruiqi Xu, Shuyuan Zheng, and Jianbin Qin. 2026. [Not all flips are conformity: Decomposing stance convergence in multi-agent LLM debate](#). *arXiv preprint arXiv:2606.00820*.
- Shiyu Ji, Farnoosh Hashemi, Joice Chen, Juanwen Pan, Weicheng Ma, Hefan Zhang, Sophia Pan, Ming Cheng, Shubham Mohole, Saeed Hassanpour, Soroush Vosoughi, and Michael Macy. 2025. [A generalizable rhetorical strategy annotation model using LLM-based debate simulation and labelling](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 20482–20503, Suzhou, China. Association for Computational Linguistics.
- Sanjeev Manivannan. 2026. [BoardroomAI: Dependency-aware human-steerable multi-agent deliberation through evolving decision graphs](#). *arXiv preprint arXiv:2608.13046*.

- Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. [Generative agents: Interactive simulators of human behavior](#). In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.
- Francesco Salvi, Manoel Horta Ribeiro, Riccardo Galletti, and Robert West. 2025. [On the conversational persuasiveness of GPT-4](#). *Nature Human Behaviour*.
- Henning Wachsmuth, Martin Potthast, Khalid Al Khatib, Yamen Ajjour, Jana Puschmann, Jiani Qu, Jonas Dorsch, Viorel Morari, Janek Bevendorff, and Benno Stein. 2017. [Building an argument search engine for the web](#). In *Proceedings of the 4th Workshop on Argument Mining*, pages 49–59, Copenhagen, Denmark. Association for Computational Linguistics.
- Joshua C. Yang, Maurice Flechtner, Damian Dailisan, and Michiel A. Bakker. 2026. [Belief engine: Configurable and inspectable stance dynamics in multi-agent LLM deliberation](#). *arXiv preprint arXiv:2605.15343*.
- Joshua C. Yang, Marcin Korecki, Damian Dailisan, Carina I. Hausladen, and Dirk Helbing. 2024. LLM voting: Human choices and AI collective decision making. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, volume 7, pages 1696–1708.